PARUL TANDON, *Mother of* **SAHANA**, *7, Bengaluru*

Explanation goes a long way in sensitising people, feels Tandon, a home baker and former HR professional. "I have another daughter who is not autistic. I have explained to her why Sahana needs a bit more of my attention. I was honest about her special needs," she says. Today, her elder sister and their friends have taken on the responsibility of looking after Sahana. "It took me almost two years to make them understand.... They accept her fully because children are naturally empathetic," says Tandon.

When Sahana was initially diagnosed, it was heartbreaking but the family never thought to hide it or feel ashamed. Instead, they started building their life around her needs. Holidays are planned around Sahana's therapy classes. "There is more information about autism today because of the internet, but as parents the initial few years can still be confusing and overwhelming. It helps to have support groups and family," says Tandon, who has retired from full-time work to devote herself to Sahana. "Sahana learned to take a bath on her own at seven. People take such small milestones for granted. For us, it was a lot of work and brings a lot of joy," she says.

damagingly, not "smart". Partly because of the historically more strictly regulated socialisation of girls—with them actually required to be 'shy', as a virtue—its occurrence tended to be masked. Though neural differences among the genders are also posited, researchers have often cited plain old bias as a likely factor in the huge disparity in diagnosis. Even present-day figures in the US reckon the frequency of autism at one in every 42 males and one in every 189 females. But in either case, what causes it?

Today, scientists know that the way neurons in an autistic brain connect and respond to each other is different from how they do in a neurotypical brain. What they haven't been able to determine is exactly what these differences are because of the sheer profusion of the neural processes involved in speech, socialisation and behaviour patterns and the fact that our brain has over 100 billion neurons. One theory is that during brain development in the first two years, nerve cells take an atypical course when they are developing and branching out. That can lead to "impairment in the areas of speech or behaviour", says Dr Jigyasha Sinha, consultant paediatric neurologist and epileptologist at the Kokilaben Dhirubhai Ambani Hospital in Mumbai.

A few things, however, are becoming clear. First, that autism is largely a genetic disorder. But that does not explain why parents or siblings do not have the condition. As Dr P.S. Narang, director of paediatrics at Max Hospital in Saket, Delhi, explains, "You can have a predisposition towards a disorder, but environmental factors also play a role. Till date, we have blamed everything under the sun for autism, but the truth is, we don't have the answer. The best we have is that it is a combination of some genes and environmental stimulation that causes autism. One without the other may or may not present itself as autism."

A landmark 2020 study in the journal *Science Direct* implicated 102 genes for ASD. But most of these genes were de novo variants (seen for the first time), or genes present in an individual but absent in their parents. Then, in 2022, the largest sequencing study of ASD published in *Nature Genetics* identified four genes in autistic persons that are also present in parents. In either case, environmental factors triggered the neurological impairment in individuals with a genetic predisposition leading to autism. "We know some risk factors that precipitate it—early toxic exposure, infections during pregnancy, poor maternal health or mental state during the last trimester," says Dr Sinha.

An analysis of 84,000 mother-child autistics published in the American medical journal *Jama Pediatrics* in 2022 highlighted another strong risk factor for ASD—screen time. It showed that among boys, longer screen time exposure—television, computers, phones—at age one was significantly associated with the emergence of autism at age three. Boys seemed to be three times more at risk than girls. "Screen time is being increasingly linked to autism. It isn't uncommon for parents to show babies rhymes or cartoons on a phone," says Dr Veena Kalra, paediatric neurologist at Apollo Hospitals, Delhi, who sees around 20 autistic children daily. "A young brain is not used to processing so much information and it results in screen fixation, information overload, hyperactivity and possibly damage to how the brain begins to process speech and communication."

A few small-sample studies have looked at other factors, such as the age of the parents. A 2016 study published in *PubMed* showed that women at the start or end of their childbearing age are more likely to have autistic children. According to a 2006 study in JAMA Network, men who become fathers after 40 are six times more likely to have an autistic child than men who father kids before 30. "We have some cause to worry because these days, people want to have children at a later age. It is no wonder that we are seeing more children develop the condition than we did a decade ago," says Dr Narang.

IN ADDITION TO GENETIC PREDISPOSITION, ENVIRONMENTAL FACTORS ARE BEING SEEN AS A TRIGGER FOR AUTISM

In the US, the number of diagnosed autistic persons has nearly tripled since the 2000s. In India, however, there is no census on invisible disabilities like autism. A 2018 study by INCLEN Trust International pegged it to be one in 100 children under the age of 10 in India. A 2019 study by the Indian Council of Medical Research (ICMR) in the journal *Neurology India* cited the same figure, but noted that high-quality larger studies are an urgent need to understand the burden of autism.